

Tzu-Kun Hsiao

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EDUCATION

PhD in Information Sciences

2026

University of Illinois Urbana-Champaign, IL, USA

Dissertation title: "Semantic Citation: Mentions of Knowledge in Citation Contexts"

Advisor: Dr. Vetle I. Torvik

Committee: Dr. Halil Kilicoglu, Dr. Jodi Schneider, Dr. Matthew Turk

MA in Library & Information Science

2017

National Taiwan University, Taipei, Taiwan

Thesis title: "Diversity of Forward Citation of Highly Cited Researchers in the Field of Chemistry and Materials Science"

Advisor: Dr. Mu-Hsuan Huang

BA in Library & Information Science

2013

National Taiwan University, Taipei, Taiwan

WORK EXPERIENCE

Assistant Professor

Summer 2026 - present

Scholar Support and Data Services, University of Mississippi Libraries

RESEARCH EXPERIENCE

Doctoral Researcher

Fall 2018 - Spring 2026

School of Information Sciences, University of Illinois Urbana-Champaign

- Develop a method integrating bibliometrics, machine learning, and natural language processing to assess research impact through the diversity of knowledge use (current project).
- Built a pipeline to systematically identify citation contexts from 2 million articles and deposited the resulting dataset, OpCitance, in the Illinois Data Bank to support open science and reuse.
- Designed a method for detecting near-duplicate records in the Semantic Scholar Open Research Corpus (S2ORC) to improve data quality for downstream analyses.
- Modeled temporal patterns of knowledge transfer within and across patents and papers.

Research Assistant

Fall 2020 - Spring 2021

School of Information Sciences, University of Illinois Urbana-Champaign

Supervisor: Dr. Jodi Schneider

- Curated and analyzed bibliographic and citation data for the Reducing the Inadvertent Spread of Retracted Science (RISRS) project.
- Investigated the diffusion of retracted articles and appropriateness of post-retraction citations using bibliometrics, sentiment analysis, and content analysis.
- Co-evaluated a filtering strategy that reduced workload in systematic review preparation.

Research Assistant

Fall 2018 - Spring 2019

School of Information Sciences, University of Illinois Urbana-Champaign

Supervisor: Dr. Vetle I. Torvik

- Conducted research on knowledge transfer between science and technology.

Research Assistant

Fall 2014 - Spring 2018

Department of Library and Information Science, National Taiwan University

Supervisor: Dr. Mu-Hsuan Huang

- Collected and cleaned data for building institutional profiles and compared university ranking methodologies to support evidence-based assessment.

TEACHING EXPERIENCE

Co-Instructor & Co-Developer of IS597 Research Impact

Spring 2023

School of Information Sciences, University of Illinois Urbana-Champaign

- Co-developed and co-instructed this new graduate-level course with Professor Jodi Schneider, including syllabus design and weekly reading selection.
- Designed and led in-class lab sessions on data analysis using Python and VOSviewer.
- Delivered lectures and led discussions on bibliographic databases, research impact indicators, and research profiles.
- Created two assignments for students to gain hands-on experience analyzing bibliographic data and constructing research profiles.
- Created the final project and scheduled regular milestone meetings with each student to provide guidance and feedback on their final project.
- Awarded “Teachers Ranked as Excellent by Their Students”.

Co-instructor of IS445 Data Visualization

Fall 2022

School of Information Sciences, University of Illinois Urbana-Champaign

- Delivered lectures and led discussions on data preparation, visualization types, and interactive visualizations.
- Developed and taught lab sessions using Python packages, including *matplotlib*, *ipywidgets*, *bqplot*, and *igraph*.

Teaching Assistant

Fall 2019 - Spring 2025

School of Information Sciences, University of Illinois Urbana-Champaign

- IS445 Data Visualization (Spring 2025, Spring & Fall 2024, Fall 2023, Spring 2022, Fall 2021):
 - Taught a section on network visualizations, including hands-on demos with *bqplot* and *igraph*.
 - Developed grading rubrics, graded assignments, and managed online learning systems (Canvas and PrairieLearn).
- IS205 Programming for Information Problems (Summer 2023); IS430 Foundations of Information Processing (Summer 2022); PR590 Programming for Analytics and Data Processing (Spring 2020):
 - Assisted students with Python programming problems, and graded assignments.
- IS457 Introduction to Data Science (Fall 2019):
 - Planned an assignment using R for statistical analysis and graded assignments.

PUBLICATIONS

Peer-reviewed Journal Articles

- Hsiao, T.-K., & Torvik, V. I. (2023). OpCitance: Citation contexts identified from the PubMed Central open access articles. *Scientific Data*, 10, 243. <https://doi.org/10.1038/s41597-023-02134-x>
- Hsiao, T.-K., & Schneider, J. (2021). Continued use of retracted papers: Temporal trends in citations and (lack of) awareness of retractions shown in citation contexts in biomedicine. *Quantitative Science Studies*, 2(4), 1144–1169. https://doi.org/10.1162/qss_a_00155

Peer-reviewed Conference Proceedings

- Chen, P.-Y., Hsiao, T.-K. & Sugimoto, C. R. (2025). Exploring the policy impact and funding mechanisms of scientific collaboration between Taiwan and New Southbound Policy (NSP) priority countries. *Proceedings of ISSI 2025 – the 20th International Conference on Scientometrics & Informetrics*, 2, 2318–2145. https://doi.org/10.51408/issi2025_103
- Hsiao, T.-K. (2023). Appropriateness of citing retracted articles in biomedicine: Sentiments expressed in citations without acknowledgement of retraction. *Proceedings of ISSI 2023 – the 19th International Conference of the International Society for Scientometrics and Informetrics*, 2, 181–187. <https://doi.org/10.5281/zenodo.8370893>
- Hsiao, T.-K. (2023). Document conflation of a large scholarly full-text dataset. *Proceedings of the Association for Information Science and Technology*, 60(1), 977–979. <https://doi.org/10.1002/pra2.917>
- Proescholdt, R., Hsiao, T.-K., Schneider, J., Cohen, A. M., McDonagh, M. S., & Smalheiser, N. R. (2022). Testing a filtering strategy for systematic reviews: Evaluating work savings and recall. *AMIA Annual Symposium Proceedings*, 2022, 406. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9285169/>
- Hsiao, T.-K., Fu, Y., & Schneider, J. (2020). Visualizing evidence-based disagreement over time: The landscape of a

public health controversy 2002–2014. *Proceedings of the Association for Information Science and Technology*, 57(1), e315. <https://doi.org/10.1002/pr2.315>

Hsiao, T.-K., & Torvik, V. I. (2020). Technology footprints in scientific discovery: Citation contexts of paper-to-patent citations. *Proceedings of the Association for Information Science and Technology*, 57(1), e337. <https://doi.org/10.1002/pr2.337>

Hsiao, T.-K., & Torvik, V. I. (2019). Knowledge transfer from technology to science: The longevity of paper-to-patent citations. *Proceedings of the Association for Information Science and Technology*, 56(1), 417–421. <https://doi.org/10.1002/pr2.41>

Dinh, L., Sarol, J., Cheng, Y.-Y., Hsiao, T.-K., Parulian, N., & Schneider, J. (2019). Systematic examination of pre-and post-retraction citations. *Proceedings of the Association for Information Science and Technology*, 56(1), 390–394. <https://doi.org/10.1002/pr2.35>

Cheng, Y.-Y., Parulian, N., Hsiao, T.-K., Dinh, L., Sarol, J., & Schneider, J. (2019). ReTracker: Actively and automatically matching retraction metadata in Zotero. *Proceedings of the Association for Information Science and Technology*, 56(1), 372–376. <https://doi.org/10.1002/pr2.32>

Tang, M.-C., Hsiao, T.-K., & Ou, I.-A. (2017). Not all books in the user profile are created equal: Measuring the preference “representativeness” of books in aNobii online bookshelves. *Proceedings of the International Conference on HCI in Business, Government, and Organizations*, 424–433. https://doi.org/10.1007/978-3-319-58484-3_33

OTHER CONFERENCE PRESENTATIONS

Hsiao, T.-K., & Torvik, V. I. (2026, June 29-July 1). *Tracing Knowledge Diffusion by Citation Context Embeddings*. [Poster]. 5th International Conference on the Science of Science & Innovation, Boulder, United States.

Hsiao, T.-K., (2025, November 14-18). *Semantic citation: Mentions of knowledge in citation contexts*. [Oral presentation]. Doctoral Colloquium at 88th Association for Information Science and Technology (ASIS&T) Annual Meeting, Virginia, United States.

Chen, P.-Y., & Hsiao, T.-K. (2024, July 18-20). *Exploring international collaboration dynamics through multilayer network analysis*. [Oral presentation]. 10th International Conference on Computational Social Science, Philadelphia, PA, United States. <https://www.ideals.illinois.edu/items/135974>

PROFESSIONAL SERVICE

- Reviewer for journals including *Scientific Data*, *Scientific Reports*, and *Journal of Academic Ethics*
- Reviewer for the *Annual Meeting of the Association for Information Science and Technology (ASIS&T)*

HONORS & AWARDS

Berner-Nash Memorial Award	2026
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Doctoral Dissertation Award

List of Teachers Ranked as Excellent by Their Students	2023
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Awarded based on outstanding student evaluations in teaching.

Phi Tau Phi Scholarly Honor	2017
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Awarded to outstanding students selected as departmental representatives.

TECHNICAL SKILLS

Programming & Data Analysis

- Python: Experienced in natural language processing, machine learning, statistical analysis, and data visualization.
- MySQL: Management and retrieval of large scale bibliographic data.

Online Assessment & Learning Systems

- PrairieLearn and Canvas: Maintained and managed course sites, including content updates, deploying assignments, and troubleshooting technical issues.

Research Databases

- Proficient in curating data from bibliographic databases including Web of Science, Scopus, and Dimensions AI.